

Topics Covered:

- Data organization
- Data presentation
- Data summarization I: Measure of central tendency
- Data summarization II: Measure of variation

Lab Question

1. Write a python code to construct frequency distribution table that contains the following columns (class limits, class boundaries, class frequency, class relative frequency, accumulated frequency and accumulated relative frequency) for the following data that represents the high temperature in Fahrenheit of 50 states in US: *using 7 classes*

112, 110, 107, 116, 120, 100, 118, 112, 108, 113, 127, 117, 114, 110, 120, 120, 116, 115, 121, 117, 134, 118, 118, 113, 105, 118, 122, 117, 120, 110, 105, 114, 118, 119, 118, 110, 114, 122, 111, 112, 109, 105, 106, 104, 114, 112, 109, 110, 111, 114.

Then use the output to draw *histogram, frequency polygon, and ogive plot* for the class frequencies. Also Calculate the *mean and variance* for both raw data and grouped data. Calculate the variance when the data is considered as a *population* and when it is considered as a *sample*.

Assignment Question

Using the given data in Q. 1 to draw *histogram, frequency polygon, and ogive plot* for the class *relative* frequencies.

